

Procurement Management	Software Requirement Specification
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→ **Revision History:**

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1. Introduction

1.1 Purpose

This document outlines the software requirements for the Procurement Management System under the Youth Skill Development Training Program. The system aims to digitize and streamline supplier registration, RFP (Request for Purchase) creation, quotation submission, and evaluation. It ensures efficient workflows, automated decision-making, and structured approval processes for suppliers, reviewers, and approvers.

1.2 Scope

The system provides a centralized platform for managing procurement activities, ensuring efficiency, transparency, and accuracy in supplier onboarding, RFP handling, and quotation evaluation. Key functionalities include:

1. **Supplier Registration:** Secure onboarding with OTP verification and document submission.
2. **RFP Management:** Reviewers create and publish RFPs, track supplier bids, and manage approvals.
3. **Quotation Evaluation:** Reviewers assess supplier quotations, assign scores, and recommend the best offers.
4. **Final Approvals:** Approvers finalize supplier selection and approve/reject RFPs.
5. **Automated Reporting & Dashboards:** Real-time analytics for tracking supplier activities, RFP trends, and decision insights.

1.3 Intended Stakeholder

1. **Project Sponsors and Academic Mentors:** Representatives from BJIT Academy and related training programs who oversee the project's academic and operational outcomes. They ensure the project aligns with training objectives and provides value to the participants.
2. **End Users:** Internal users who interact with the system to carry out their tasks. These include suppliers, reviewers, approvers, and other roles engaged in the procurement process. They ensure the system is used as intended for everyday operations.

3. **Administrators:** Responsible for managing and maintaining the system's configurations, ensuring that user roles, permissions, and system security are appropriately set up and adhered to. They ensure smooth operation across the various modules.

1.4 References

Reference	Location
Odoo 17	https://www.odoo.com/documentation/17.0/developer.html
Odoo 17 Coding Guidelines	https://www.odoo.com/documentation/17.0/contributing/development/coding_guidelines.html

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
RFP	Request for Purchase
RFQ	Request for Quotation
OWL	Odoo Web Library
OTP	One Time Password
RBAC	Role-Based Access Control
UI/UX	User Interface / User Experience
SMTP	Simple Mail Transfer Protocol
SRS	Software Requirements Specification
Nginx	A high-performance web server used as a reverse proxy for handling requests.
Docker	A containerization platform used for deploying the application in an isolated environment.
PostgreSQL	A relational database management system for storing application data.

2. Overall Description

2.1 Overview

The Procurement Management System is designed to digitize and streamline supplier onboarding, RFP (Request for Purchase) management, and quotation evaluation. It provides a structured workflow for suppliers, reviewers, and approvers, enhancing efficiency, transparency, and decision-making in procurement processes.

1. **Product Perspective:** This system is a modular extension of Odoo 17, integrating seamlessly with its procurement, user management, and reporting functionalities. It follows Odoo's best practices to ensure compatibility, scalability, and security.

2. Product Functions:

- Ensures the authenticity of supplier accounts before registration.
- Collects necessary company and financial details with field validation.
- Enables suppliers to upload legally required documents with size and format validation.
- Reviewers define product requirements, deadlines, and submission criteria.
- Approvers reviews RFPs before making them available for supplier bids.
- Suppliers submit bids while reviewers track and manage submissions.
- Suppliers provide pricing, delivery terms, and product details.
- Reviewers evaluate quotations based on cost, delivery time, and terms.
- The best-qualified suppliers are identified based on evaluation scores.
- Generates structured Excel and Qweb reports on RFPs and supplier activities.
- Displays key procurement metrics such as approved RFQs, supplier performance, and total expenditures.
- Uses graphs, charts, and tables to provide actionable insights.

3. User Characteristics:

User Role	Responsibilities
Suppliers	Register, upload documents, and participate in RFP submissions.
Reviewers	Create and manage RFPs, evaluate supplier applications, and score quotations.
Approvers	Finalize supplier selection, approve/reject RFPs, and oversee procurement processes.
Administrators	Manage system configurations, enforce security policies, and maintain user access control.

4. Constraints:

- The system must strictly follow Odoo 17's modular architecture to ensure maintainability.
- The dashboard and interactive UI elements must be developed using OWL (Odoo Web Library).
- The system must be Dockerized, with Nginx as a reverse proxy for optimized performance.
- Role-Based Access Control (RBAC) must be enforced, limiting data access based on user roles.

5. Assumptions and Dependencies:

- The system assumes a fully functional Odoo 17 environment, including PostgreSQL as the backend database.

- OTP verification and notifications depend on a configured SMTP server for email delivery.
- The system requires Linux-based hosting (Ubuntu preferred) with sufficient resources for optimal performance.
- The procurement process must comply with regulatory policies related to document validation and data retention.

2.2 Technical Platform

The Procurement Management System will be developed using Odoo 17 with a modular architecture, ensuring maintainability, scalability, and adherence to best practices.

1. Backend Technologies:

- Framework: Odoo 17
- Programming Language: Python
- Database: PostgreSQL
- ORM: Odoo ORM for secure and optimized database interactions
- Authentication: Odoo's built-in authentication with OTP-based email verification
- Business Logic: Implemented within Odoo models (.py files) using Python

2. Frontend Technologies:

- Framework: OWL (Odoo Web Library) for interactive UI components
- UI Components: XML-based views (Form, List, Kanban, and Graph Views)
- Styling: Bootstrap
- Event Handling: OWL state management for real-time updates in dashboards and reports

3. Deployment & Infrastructure:

- Containerization: Docker-based deployment for portability and dependency management
- Web Server: Nginx as a reverse proxy for load balancing and security
- Multi-Container Setup:

- Odoo Backend (Application logic and processing)
 - PostgreSQL Database (Data storage)
 - Nginx Proxy (Handles requests and optimizes performance)
- Environment Configuration: Docker Compose for multi-container orchestration

4. Database Management:

- Database Engine: PostgreSQL
- Data Relationships: Managed via Odoo's ORM with foreign key constraints
- Indexes & Optimization: Optimized indexing for faster queries on RFPs, quotations, and supplier data
- Backup Strategy: Regular automated backups to prevent data loss

5. Operating System & Hosting:

- Operating System: Linux (Ubuntu 20.04+ recommended)

3. Functional Requirements

3.1 Overview

Serial No	Main Features	Description
1	Email Verification	OTP-based supplier account creation
2	Supplier Registration	Multi-step form submission with document upload
3	Two-Step Verification	Review and approval process for supplier onboarding
4	RFP Management	Creation, publication, and quotation submission
5	Reporting	Generate supplier-wise RFP reports
6	Data Visualization	Graph and pivot views for RFP insights
7	Dashboard	Interactive OWL-based data analytics

3.1.1 Feature 1 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Email Validation	Check if the email is already registered or blacklisted. If either condition is true, display an error.	Temporary email services may bypass validation if not explicitly blacklisted.	
Procurement Management_002	OTP Generation	If the email is valid, send an OTP to the email.	If the user provides an incorrect email address, OTPs may be sent to unintended recipients.	
Procurement Management	OTP Input	After receiving the OTP, the user will input it in a provided field.	If a user delays inputting the OTP, it may expire before submission.	

nt_003				
Procurement Management_004	OTP Validation	If invalid or expired, display an error. If valid, proceed to the registration form with a success message.	If multiple OTPs are generated quickly, users might enter an outdated one, leading to validation failure.	

3.1.2 Feature 2 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Registration Form Sections	The form has 5 sections: Company Information, Financial Information, Client References, Certifications, and Document Submission.		
Procurement Management_002	Field Validation	Ensure mandatory fields are filled, numeric fields, date fields, optional fields follow the correct format (e.g., alphanumeric TIN), and documents are within the 1MB size limit.	No validation against auto filler extensions.	
Procurement Management_003	Confirmation and Submission	After successful validation, submit the form and send email notifications to assigned reviewers. Also, The system should provide an error message if validation fails, specifying the issues.		

3.1.3 Feature 3 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
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Procurement Management_001	First Review (Reviewer Group)	Reviewers assess the supplier application, with options to approve, reject, or blacklist. Reviewers can add comments for the reason of rejection or blacklist..		
Procurement Management_002	Final Approval (Approver Group)	The approver reviews the decision and either approves or rejects the application.		
Procurement Management_003	Vendor Creation	After final approval, create a vendor record and supplier user account.		
Procurement Management_004	UI Views	Include Form, Tree, and Kanban views for tracking, with search and filter functionality.		

3.1.4 Feature 4 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	RFP Creation by Reviewer	Only reviewers can create RFPs, which are linked to RFQs submitted by suppliers. Each reviewer can view only their own RFPs.		
Procurement Management_002	RFP Publication	After creating an RFP, the reviewer can submit it for approval. The system sends email notifications to the approver and the suppliers when an RFP is published.	After submitting RFP reviewers can't edit it.	
Procurement	Quotation Submission by	Suppliers submit quotations with pricing and delivery details. An RFQ is created		

Management_003	Supplier	and linked to the RFP.		
Procurement Management_004	Quotation Review by Reviewer	Reviewers evaluate quotations, scoring based on criteria (e.g., delivery date, charges), and can recommend suppliers.		
Procurement Management_005	Quotation Acceptance by Approver	Based on the recommended quotations by the reviewers, Approver can choose a single quotation and accept it.		

3.1.5 Feature 5 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Report Generation	A menu will allow users to generate RFP reports based on a specific supplier and date range.		
Procurement Management_002	Report Content	The approver reviews the decision and either approves or rejects the application.		
Procurement Management_003	Report Format	The report model will include two buttons: a Qweb-HTML Preview that generates an HTML preview of the report and an Excel Report that generates an Excel report using the 'xlsxwriter' package.		
Procurement Management_004	Report Validation	The start date must be earlier than or equal to the end date. The supplier must have accepted RFPs to generate a valid report.		

3.1.6 Feature 6 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Graph View	The default graph view should display RFP data on a line graph with the required date as the x-axis (monthly basis) and the total amount as the y-axis. The legend should display RFP number, supplier, and status.		
Procurement Management_002	Pivot View	The pivot table should display supplier data by month, including RFP status and total amounts. The system should display the data in descending order of total amounts.		

3.1.7 Feature 7 - Requirements:

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Supplier and Date Range Selection	Users can select a supplier from a dropdown list. Users can specify a date range (e.g., this week, last month) to filter the data.		
Procurement Management_002	Key Metrics Display	The dashboard should display key metrics for the selected supplier: Approved RFQs, Total Amount and Product-Wise Breakdown.		
Procurement	Interactivity	The dashboard must be dynamic	State management using OWL	

nt Managemant_003		and interactive, allowing users to change the supplier and date range..	could become inefficient for more complex dashboard components, causing UI slowdowns or inconsistent states in dynamic data updates.	
Procurement Managemant_004	Performance and Responsiveness	The dashboard should load quickly and be responsive across devices, using OWL for state management.	OWL's state management might not scale as expected with complex interactions and multiple live data updates, potentially affecting the overall responsiveness.	

3.1.8 Feature 8 - Deliverables:

1. **Docker Containerization:** Package the entire application with dependencies and configurations to ensure smooth operation in a Docker container.
2. **Nginx Configuration:** Use Nginx as a reverse proxy for efficient traffic routing, load balancing, and SSL termination.
3. **Documentation:** Provide clear dependency lists, installation guides, and detailed steps for deploying and troubleshooting the Dockerized application.

4. User Interface

UI No.	UI Name	Description	Test Case Identifiers
Procurement_management_001	Supplier Portal	Allows suppliers to register and submit documents	
Procurement_management_003	Reviewer Panel	Enables reviewers to evaluate applications and manage RFPs	
Procurement_management_003	Approver Panel	Enables approvers to review and finalize supplier selection	
Procurement_management_004	Dashboard	Provides key insights through data visualization	
Procurement_management_005	RFQ Submission Portal	Allows supplier to submit RFQs based on approved RFPs	

5. Non-Functional Requirements

5.1 Performance Requirements

- **Response Time:** The system should provide a response time of less than 3 seconds for any user action, such as submitting forms, loading pages, or generating reports.
- **Load Handling:** The system should be able to handle at least 100 concurrent users without degradation in performance.
- **Data Processing Speed:** Report generation and data visualization (graphs, pivot views) should process and display results within 10 seconds for standard queries.

5.2 Scalability

- The system must be scalable to support growing amounts of data, including suppliers, RFPs, and RFQs.
- The architecture should allow easy addition of new features or modules in the future, such as additional report types or new data visualizations.

5.3 Usability

- **User Interface:** The system must provide a clean, intuitive, and responsive interface. It should adhere to Odoo's UI/UX guidelines and be compatible with various screen sizes (desktop, mobile).
- **Error Messages:** All error messages should be clear and provide actionable steps for the user to correct the issue (e.g., "Invalid email format," "File size exceeded the 1MB limit").
- **Help and Support:** The system should provide an integrated help section for users to understand the registration and quotation submission process.

5.4 Availability

- The system should be available 99.5% of the time, with planned maintenance windows not exceeding 4 hours per month.

- If the system is unavailable, users should receive appropriate error messages indicating that the service is temporarily down.

5.5 Security

- **Access Control:** The system must enforce role-based access control, ensuring that users only access data and perform actions permitted by their roles.
- **Audit Trail:** Every action (e.g., submission, approval, rejection) performed on the system should be logged with details of the user and timestamp for auditing purposes.

5.6 Maintainability

- **Code Documentation:** All code should be properly commented and documented to ensure that it is easy for developers to understand, extend, and maintain in the future.
- **Error Handling:** The system should log errors clearly, with sufficient context, and send alerts to administrators when critical issues arise. The logs should be easily accessible and stored in a centralized location.
- **Modularity:** The system design should be modular, allowing individual features or components (e.g., RFP creation, report generation) to be maintained and updated independently.

5.7 Compatibility

- **Browser Compatibility:** The system must be compatible with the latest versions of major browsers (Chrome, Firefox, Safari, and Edge).
- **Mobile Compatibility:** The user interface must be responsive and work seamlessly on both Android and iOS devices.

5.8 Localization

- **Language Support:** The system should support multi-language capabilities, starting with English and the option to add other languages in the future.

- **Time Zone Management:** The system should handle time zone differences correctly for events like RFQ deadlines or report generation, ensuring accurate time representations.

5.9 Localization

- **Data Encryption:** All sensitive data must be encrypted in transit and at rest using industry-standard encryption methods (e.g., TLS, AES-256).
- **Role-Based Access Control (RBAC):** Enforce strict access control based on user roles to limit access to sensitive data and functionalities.
- **Regular Security Audits:** Conduct periodic security audits and implement real-time monitoring to detect and mitigate security threats.

6. Design Constraints

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Odoo 17 Modular Structure	The system must separate models, views, controllers, and business logic following Odoo 17's architecture.	Some complex module dependencies may require manual adjustments when extending functionalities.	
Procurement Management_002	Frontend with OWL (Odoo Web Library)	Dashboards and reports must use OWL for interactivity, state management, and reactivity.	Some Odoo default views do not support OWL, requiring custom development for compatibility.	
Procurement Management_003	Responsive UI	The UI must work seamlessly on desktop and mobile devices with adaptive layouts.	Some advanced UI elements may not render correctly on all screen sizes, requiring additional CSS fixes.	
Procurement Management_004	Docker Containerization	The application must be Dockerized for easy deployment, ensuring encapsulated dependencies.	Running multiple containers may increase resource consumption, requiring optimization.	
Procurement Management_005	Nginx as Reverse Proxy	Nginx must efficiently handle HTTP requests and route traffic to the backend.	Incorrect Nginx configurations may cause deployment issues, requiring manual debugging.	
Procurement Management_006	Docker Compose for Multi-Container Setup	Odoo backend and database must be managed using Docker Compose for modular deployment.	Requires additional configuration for networking between containers and persistent data storage.	
Procurement Management_007	Odoo 17 Coding Guidelines Compliance	Code must follow Odoo's best practices, ensuring maintainability and easy debugging.	Future Odoo updates may deprecate certain methods, requiring adjustments.	
Procurement Management	Role-Based Access Control	Enforce user roles (Supplier, Reviewer, Approver, Admin) with controlled data access.	Granular permission settings may require additional administrative configurations.	

nt_008	(RBAC)			
Procurement Management_009	Scalability for Future Enhancements	The system must support additional features without major architectural changes.	Scalability may depend on infrastructure upgrades, such as database optimizations and load balancing.	

7. Software Quality Attributes

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Performance	System must respond within 3 seconds for key operations such as form submissions, report generation, and dashboard updates.	Performance may degrade under heavy concurrent loads exceeding 100 users, especially during large-scale report generation.	
Procurement Management_002	Security	The role-based access control (RBAC) enforced must be implemented.	RBAC restrictions may limit flexibility for users requiring cross-functional access, necessitating manual overrides by administrators.	
Procurement Management_003	Scalability	The system should support at least 100 concurrent users without performance degradation.	Additional scaling may require load balancing, database optimization, and increased server resources, which could increase infrastructure costs.	
Procurement Management_004	Usability	UI must be intuitive, responsive, and mobile-friendly, following Odoo 17's UX guidelines.	Unplanned downtime due to server failures, updates, or maintenance may temporarily disrupt operations.	
Procurement Management_005	Reliability	System availability must be 99.5% uptime, with automated error handling and logging.	Unplanned downtime due to server failures, updates, or maintenance may temporarily disrupt operations.	
Procurement Management_006	Maintainability	Code must follow Odoo 17 best practices, be modular, and allow easy updates and debugging.	Future Odoo version upgrades may introduce breaking changes, requiring system modifications for compatibility.	

8. Other Requirements

Requirement ID	Description	Acceptability Criteria	Limitations	Test Case Identifiers
Procurement Management_001	Exportable Reports	Reports must be exportable in Excel format, ensuring structured data presentation.	Excel compatibility issues may arise if different versions of Microsoft Excel or alternative spreadsheet software are used.	
Procurement Management_002	Email & Notifications	Notifications must be triggered via email and dashboard alerts for key actions (e.g., RFP approval, supplier registration).	Dashboard notifications depend on real-time updates, which may introduce minor delays due to system processing.	
Procurement Management_003	System Logs	System logs must be maintained for critical actions, including supplier registrations, RFP approvals, and quotation evaluations, ensuring traceability and auditability.	Log storage can grow significantly, requiring periodic cleanup and archiving to maintain system performance.	